From Price Floors to Strategic Processing Capacity Guarantees

A proposal for the Agreement on Trade in Critical Minerals (ATCM)

To: Office of the United States Trade Representative
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Origin of this Proposal

I did not begin this work by following U.S. critical-minerals policy. I took a current systems problem — the structural inability of market mechanisms to sustain alternative processing capacity — and, over the course of one week with analytical support from AI systems, developed an independent architecture. That architecture turned out to be close to the direction U.S. policy has itself been moving (price mechanisms, investment certainty, allied coordination). However, the solution contains several elements that current policy has not yet addressed. Those missing elements are the reason for this letter.

The proposal is therefore not derived from existing U.S. policy. It was developed independently. The strong correlation with the emerging ATCM architecture is convenient, but the intellectual origin is separate.

1. What Is Happening — Verifiable Facts

Fact 1. China is the leading refiner for 19 of 20 strategic minerals tracked by the IEA, with an average market share of approximately 70%. For rare earths and graphite the share remains near or above 85–90% (IEA Global Critical Minerals Outlook).

Fact 2. Diversification of mining has outpaced the creation of alternative processing capacity. Outside the dominant supplier, utilization of smelting/refining capacity has been structurally lower, while capital and operating costs for new non-Chinese processing facilities remain substantially higher.

Fact 3. USTR is actively designing a plurilateral Agreement on Trade in Critical Minerals (ATCM) that contemplates border-adjusted price mechanisms / price floors as a way to restore investment certainty among like-minded partners (USTR Federal Register notice, February 2026; subsequent Action Plans).

Fact 4. Price-floor arrangements have already begun to appear in specific U.S. support deals. These instruments improve project economics but do not yet define how much alternative capacity must remain available, how quickly it must be able to replace a dominant external supplier, or how the strategic value of maintaining that capacity should be measured and paid for.

Fact 5. The core vulnerability is therefore not simply “insufficient mining.” It is the economic non-reproducibility of reserve processing capacity under ordinary market conditions.

2. Systemic Error — One Hypothesis

Hypothesis: Markets correctly price the product (the mineral or intermediate). They do not adequately price the strategic option value of maintaining alternative processing capacity that can be activated when the dominant external supplier becomes unavailable or coercive. As a result, capacity that is systemically necessary for resilience remains commercially vulnerable whenever globally traded material is cheaper.

Interpretation: A pure price-floor approach can make an individual project bankable. It does not, by itself, establish a measurable requirement for how much alternative capacity must exist, how fast it must be able to scale, or how its continued availability should be valued as a distinct economic object.

3. Proposed Mechanism: Strategic Capacity Guarantee (SCG)

Convert the emerging price mechanism from an instrument that supports individual mineral projects into an instrument that guarantees the continued economic viability of a defined quantity of strategic processing capacity.

Core commitment: Governments and/or qualified industrial buyers commit to long-term purchase or availability payments for the strategically required capacity. The state (and aligned buyers) create a decades-long guaranteed market and demand for those who build or maintain the technological chain needed for critical materials. This is not a short-term offtake. It is a structural demand signal measured in decades.

Under an SCG the public and allied side purchases not only the mineral, but the right to have an alternative. The economic object becomes: *qualified, available, and activatable processing capacity measured by Time-to-Replace*.

The guarantee is two-sided: the state (or qualified buyers) guarantee the producer a market; the producer guarantees the state the availability of capacity. This is a capacity contract. When external prices temporarily collapse, the facility does not have to close — the strategic optionality is paid for explicitly.

Time-to-Replace (TtR) — the time required to replace the largest external dependency in a critical industrial chain with qualified alternative capacity under defined conditions. TtR turns resilience from a qualitative aspiration into a measurable design parameter.

Conceptual value equation: $V_{strategic} = V_{product} + V_{resilience} + V_{optionality}$

4. Who Must Do What

USTR: Incorporate SCG language, long-term availability payments, and TtR metrics into the design of ATCM price mechanisms and related Action Plans; require that supported capacity carry explicit availability and activation obligations.

Allied governments / qualified buyers: Commit to long-term (multi-decade) purchase or availability payments that create a durable demand signal for strategic capacity; align offtake, stockpiling and industrial-policy instruments.

Industry (processors & downstream): Commit to maintain defined readiness levels and accept transparency on cost structures in exchange for long-term economic certainty.

Finance / DFIs / Export credit: Treat SCG-backed capacity as a distinct risk category with improved bankability once availability obligations are contractually defined.

Independent technical function: Define qualification standards, measure TtR, and verify continued availability — preventing the mechanism from becoming a pure subsidy.

5. Why Participants Will Agree — Specific Interests

USTR / U.S. Government: Gains a concrete, measurable layer that turns price floors into a tool for systemic resilience rather than project-by-project support; strengthens the ATCM's value proposition to partners.

Allied governments: Receive a shared metric (TtR) and a shared economic object that reduces free-riding and coordination failure; gain a multi-polar system in which no single participant can economically stop the others.

Processors: Obtain long-term revenue certainty that covers the cost of maintaining strategic (not merely commercial) capacity.

Downstream manufacturers & defense: Gain a quantifiable reduction in supply-chain fragility and a clearer basis for contingency planning.

Financiers: Face a clearer risk allocation: the strategic premium is paid by the public/allied side, while commercial risk remains with the operator.

6. If the Proposal Succeeds Fully

A network of mutually substitutable processing nodes emerges. No single external center can halt a critical industrial chain by withholding intermediate materials. Time-to-Replace for priority chains falls into a politically and industrially acceptable range. Investment certainty improves because the strategic premium is explicit and durable. The system becomes multi-polar: allies retain independent capability even if one major participant changes course.

7. If the Proposal Is Adopted Only Partially

Even without a complete independent chain, the system becomes less brittle: selected reserve capacities, long-term offtakes, and strategic inventories remain in place. The risk of "expensive projects without systemic resilience" is reduced because the mechanism is designed around availability and activation, not merely around project IRR.

8. What the Proposal Makes Visible

Strategic processing capacity is not a side-effect of commercial production. It is a distinct economic asset that must be deliberately maintained. Current market and policy instruments undervalue that asset. Making it visible and measurable is the first step toward protecting it.

9. The New Political Reality This Proposal Seeks to Create

No critical industrial chain should depend on whether alternative production happens to be the cheapest in the world at any given moment. The ability to replace a dominant external supplier within a defined time becomes a standing requirement of economic and national security policy. The ultimate maturity criterion: China (or any other actor) can change price, export policy, investment, technology or relations with one partner — and the remaining participants continue to function without emergency assistance from a single guarantor.

10. Chinese Attack Surface and Three-Track Adaptive Architecture

Any mechanism that creates alternative processing capacity will be contested. The response is not an endless list of counter-measures. It is a three-track architecture that closes classes of attacks by design. The goal is not “make the United States independent of China.” The goal is a multi-polar system of critical materials in which no single participant can economically stop the others — consistent with the principle that allies should not be dependent on any single center, including the United States.

Three parallel tracks

Track	Tempo	Task	Horizon
1. Shield	Fast	Prevent an external actor from collapsing the system before it matures	months → 2–3 years
2. Framework	Normal	Build an economically viable network of alternative chains	2–7 years
3. Autonomy	Slow	Remove structural dependencies: technology, equipment, skills, feedstock, standards	5–15+ years

The fast track must not try to build the entire system — it buys time. The normal track turns that time into industrial capacity. The slow track removes the structural causes of dependence.

Track 1 — Shield (buy time)

- **Reserve of time, not only material.** The key metric is months of autonomous operation across the chain (feedstock → concentrate → refined material → intermediate → critical component). Reference bands: <12 months = red; 12–24 = yellow; >24 = green (thresholds set per material and sector).
- **Reserve capacity.** Maintain distributed readiness (e.g., two plants at 70% + one able to ramp quickly) rather than a single plant at 100%.
- **Guaranteed demand as capacity contract.** Governments and/or qualified industrial buyers commit to long-term purchase or availability payments. The state pays for the strategic optionality of capacity, not merely for delivered tonnes. This is the primary defense against predatory pricing.

Track 2 — Framework (turn time into industry)

Build six independent nodes: feedstock portfolio (no single country >40–60%), processing (at least two independent routes per strategic material), equipment independence, technology (process + documentation + pilot + industrial experience), workforce / industrial knowledge base, and recycling + substitution + redesign. If any single node remains monopolized, the system is incomplete.

Track 3 — Autonomy (change the structure of dependence)

Own equipment manufacturers in multiple jurisdictions; indigenous refining / separation / purification / metallurgy / process-control technologies; material-intensity reduction, substitution and new materials; long-horizon science as option value. This track does not replace the Shield or Framework — it makes dependence progressively harder to reimpose.

Primary attack classes this architecture is designed to close

Predatory pricing / temporary market flooding; equipment, technology and reagent denial; financial and offtake pressure on allies; standards and qualification capture; secondary supply-chain disruption; and the slow economic capture of allied decision-making (“we remain partners, but begin to choose the cheaper external option”).

A compact control system can be monitored through eight state variables: source diversification (S), processing capacity (P), equipment independence (E), technology independence (T), workforce (W), capital resilience (C), demand guarantee (D), and Time-to-Replace (τ). Decision triggers move from green (continue) through yellow/orange (corrective industrial policy) to red (strategic assurance mode: reserve release, capacity activation, emergency procurement, temporary rationing).

About the Author and Why This Can Be Evaluated on Its Merits

Dmitry Kazakov — software development since 2010; independent software and systems engineering practice since 2015. Work spans software, data systems, hardware-adjacent applications, security and industrial systems. Additional experience includes building businesses and operating in complex multi-stakeholder environments.

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This proposal is offered as a systems-engineering construct, not as the product of institutional authority inside U.S. trade policy. It should be judged by the clarity of the mechanism, the measurability of Time-to-Replace, the long-term demand commitment, and the robustness of the three-track architecture — not by the formal credentials of the author.

I am not asking USTR to adopt the conclusions. I am asking whether the mechanism is sufficiently useful to merit technical discussion inside the ongoing design of the ATCM and related Action Plans.

Contact: engagement via ordinary USTR channels (engagement@ustr.eop.gov or the public comment / stakeholder processes associated with ATCM development).
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